A Novel Color Edge Detection Algorithm in RGB Color Space

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Abstract—**Since edge indicates the outline of an object and also can provides important information to separate the objects from the background or other overlapping objects, edge detection is an essential tool in image processing and computer vision. Compared with gray image, color image provides more edge information of objects. However, the current color edge detection algorithms acquired so much time to compute and they are hardly used in real-time system. In order to improve the efficiency and the performance of the color edge detection, a novel color edge detection algorithm is proposed in this paper. In the propose algorithm, an improved Kuwahara filter is used to smooth the original image first. After edge detection with each channel independently in RGB color space, an adaptive threshold selection method is applied to predict the optimum threshold value and an edge thinning algorithm is used to extract accurate edge. The proposed algorithm is applied to a lots of color images and the result show that the algorithm works well with color images in different situation and it can be used in certain real-time system too.**

Keywords-color edge detection; RGB color space; improved Kuwahara filter; adaptive threshold selection; edge thinning algorithm

I. INTRODUCTION

Edge is one of the most fundamental and important features of image. Edge detector is one of the most important tools in image processing [1]. Efficient and accurate edge detection will directly work on the understanding of machine vision system to objective world. Novak and Shafer [3] found that 90 percent of the edges are about the same in gray value and in color images. Consequently, there are still 10 percent of edge information in a color image, which can be essential in certain computer vision tasks, cannot be found in the corresponding gray image [2]. Compare with gray image which a lot of mature edge detection techniques have been proposed, edge detection in color image is far more challenging task.

Since Nevatia published his first paper about color edge detection in 1977 [7], there has been made a variety of methods on color edge detection. These methods can be classified accordance with their color space, but the more common approach is to classify them into two methods, synthetic and vector. The so-called synthetic method of color edge detection applies gray image edge detection algorithms to each of the three channels of the color space and synthesize the three results by a certain method. The vector method uses the various features of three-dimensional space vector to detect color image edge.

Adrian N. Evans etc.[5] propose a color edge detector based on vector differences. LI Xue-wei etc.[6] propose a novel perceptual model for edge detection for color image, which use Edge-preserving Gaussian Filter, predict algorithm and color gradient enhance method. Soumya Dutta etc.[2] propose an synthetic algorithm of edge detection based on the optimum threshold value predicting and edge thinning technique. Mark A. Ruzon etc.[8] use the compass operator to substitute vector quantization. SUN Hui-xian etc. [9] propose another synthetic algorithm based on HSI color space, which calculate the edge information of hue, saturation and intensity, and the final color edge information is obtained by combining the component edge information. R. D. Dony etc. [10] use RGB vector angles to detect edge for color image, which based on the calculation of the vector angle between two adjacent pixels. In this paper, a synthetic method is discussed to detect edges in color images with an improved Kuwahara filter to smooth the original image and after that an auto threshold algorithm and thinning technique are applied to generate the final edge detection image.

II. THE PROPOSED APPROACH

The proposed approach is comprised of four steps. At first, The original image is smoothed by an improved Kuwahara filter to subdue the noise without disturbing the sharpness and the edges. Secondly, an improved Sobel operator [11] which has four different directions is used to find out the edge information of each channels in RGB color space and after that a synthetic method is proposed to generate the color edge map. In the third step, image is threshold with a single threshold value which calculated by the grayscale statistical histogram of the color edge map. Finally, the detected color edges are processed with a thinning algorithm in order to get a better edge image. The approach of color edge detection presented in this paper is outlined in Fig. 1.

A. Smoothing by Improved Kuwahara Filter

The Kuwahara filter is an edge-preserving filter, which smoothes the image without disturbing the sharpness and the position of edges [12][13]. The traditional Kuwahara filter [12] can be implemented for different window shapes, the algorithm for a square window of size $H = W = 4L + 1$ where L is an integer, H is height of window and W is the width of window. However, the smallest size of square window is $H = W = 5$ and the least number of pixel in each region is nine, which means that the computational quantity of the algorithm for each pixel of signal channel image includes sixteen addition,

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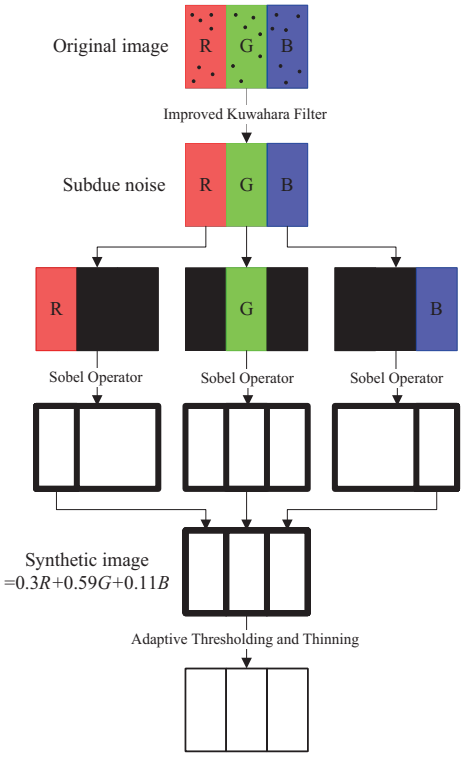


Figure 1. The flow of the propose approach.

sixteen subtraction, two division and nine multiplication at least. In order to reduce the computational quantity and solve the border effect, the improved Kuwahara filter is proposed in this paper. The window of improved Kuwahara filter is partitioned into four triangle regions as shown in Fig. 2.

In each of the four regions ($p = 1,2,3,4$), the mean brightness, m_p in (1), and the variance, s_p^2 in (2), are computed. The output value of the center pixel in the window is the mean value of that region with the smallest variance.

$$m_p = \frac{1}{\Lambda}\sum_{(x,y)\in\mathcal{R}} p(x,y) \tag{1}$$

$$s_p = \sqrt{\frac{1}{\Lambda-1}\sum_{(x,y)\in\mathcal{R}}[p(x,y)-m_p]^2} \tag{2}$$

where m_p is the mean brightness, s_p is the variance, Λ is the number of pixels within a region($\mathcal{R}$), and $p(x,y)$ is the value of the pixel with the position of (x,y).

Compare to traditional Kuwahara filter, the improved Kuwahara filter has only six pixel in each region ($\Lambda = 6$), that means only twelve addition, twelve subtraction, two division and six multiplication are needed when deal with each pixel of signal channel image. So, the computational quantity can be reduce evidently. Besides, a border processing is introduce into this filter to solve the border effect in order to get a better result.

B. Color Edge Detection by Improed Sobel Operator

The proposed color edge detection method in this paper is a synthetic method. This method separate red, green and blue channel from image first. Then improved Sobel operator is applied to each single channel image to generate edge map.

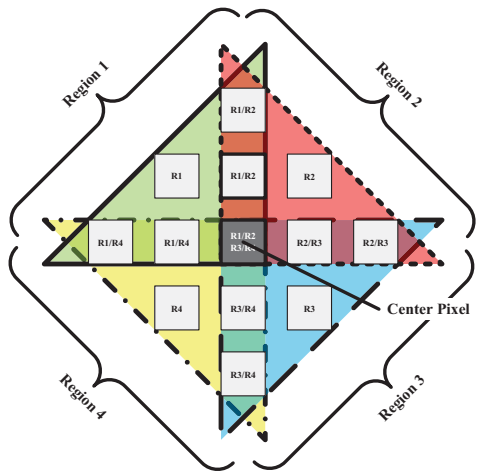


Figure 2. Four triangle regions defined for the improved Kuwahara filter.

Finally, the three edge image mention above are merge into one edge image using certain synthetic algorithm.

The difference $\Delta V(V_1, V_2)$ between two pixel is defined in (3), where V_1 and V_2 mean the value of two pixel and the range of the value lies within 0 to 255.

$$\Delta V(V_1, V_2) = |V_1 - V_2| \tag{3}$$

The improved Sobel edge detection operator [11] as shown in Fig. 3 and Fig. 4, which has four directions and they are 0°, 45°, 90° and 135°. To the pixel with the position of (x,y), $HOR(x,y)$ means the value on 0° direction of red channel. The same as $HOR(x,y)$, $NOR(x,y)$, $VOR(x,y)$ and $SOR(x,y)$ mean the value on 45°, 90° and 135° direction of red channel too. Considering $p(x,y)$ as the center pixel, the mathematical model for edge detection [9] is presented in (4), (5), (6) and (7).

$$HOR(x,y) = \left|\Delta V\left(V_{-1,-1}, V_{1,-1}\right) + 2\Delta V\left(V_{-1,0}, V_{1,0}\right) + \Delta V\left(V_{-1,1}, V_{1,1}\right)\right| \tag{4}$$

$$NOR(x,y) = \left|\Delta V\left(V_{-1,0}, V_{0,1}\right) + 2\Delta V\left(V_{-1,-1}, V_{1,1}\right) + \Delta V\left(V_{0,-1}, V_{1,0}\right)\right| \tag{5}$$

$$VOR(x,y) = \left|\Delta V\left(V_{-1,-1}, V_{-1,1}\right) + 2\Delta V\left(V_{0,-1}, V_{0,1}\right) + \Delta V\left(V_{1,-1}, V_{1,1}\right)\right| \tag{6}$$

$$SOR(x,y) = \left|\Delta V\left(V_{-1,0}, V_{0,-1}\right) + 2\Delta V\left(V_{-1,1}, V_{1,-1}\right) + \Delta V\left(V_{0,1}, V_{1,0}\right)\right| \tag{7}$$

where $V_{\mathrm{m,n}}$ means the value of the pixel with the position of $(x+m, y+n)$.

(x-1,y-1)	(x-1,y)	(x-1,y+1)
(x,y-1)	(x,y)	(x,y+1)
(x+1,y-1)	(x+1,y)	(x+1,y+1)

Figure 3. 3×3 neighborhood graph of the pixel with the positon of (x,y).

1	2	1
0	0	0
-1	-2	-1

2	1	0
1	0	-1
0	-1	-2

1	0	-1
2	0	-2
1	0	-1

0	1	2
-1	0	1
-2	-1	0

Figure 4. Four 3×3 directional masks of the improved sobel edge detection operate (0°, 45°, 90° and 135°).

The output value of the center pixel with the position of (x, y) in red channel is the max value of $HOR(x, y)$, $NOR(x, y)$, $VOR(x, y)$ and $SOR(x, y)$, which defined as $MOR(x, y)$.Also, $MOG(x, y)$ and $MOB(x, y)$ can be calculated in green and blue channel use the same method. Using the results above, the final edge value that defined as $MOE(x, y)$can be calculated now. To reduce the computational quantity, the transformed parameters of the merge synthetic algorithm have been calculated and the transformed parameters are the coefficients of three channels in (8).

$$MOE(x, y) = 0.3 \times MOR(x, y) + 0.59 \times MOG(x, y) + 0.11 \times MOB(x, y) \quad (8)$$

C. Adaptive Thresholding

Due to the existence of the weak edge information, the outline of objects in the final edge image is not so evidence. An adaptive thresholding algorithm [14] that proposed by Otsu in 1979 is used to select an optimal threshold, namely, so as to maximize the separability of the main edge and the insignificant edge information in gray levels. This method is able to select a threshold T automatically form a gray level histogram has been derived from the viewpoint of discriminate analysis.

D. Edge Thinning

There are so many thinning algorithms, including iterative deletion of pixels, nonpixel-based methods and so on [15] . It is understood that a pixel P examined for deletion is a edge pixel, and the pixels (P_1 to P_8) in its 3×3 window are labeled as shown in Fig. 5. The edge thinning algorithm proposed in this paper is one of the iterative deletion methods and it uses four groups of thinning structure elements as shown in Fig. 6 and two judging conditions as shown in (9) and (10) to decide whether delete the center pixel or not.

$$J_1 : N_p = \sum_{i=1}^{8} P_i \text{ and } N_p < 7 \text{ and } N_p > 1 \quad (9)$$

$$J_2 : T_p = 1 \quad (10)$$

where N_p means the number of nonzero points around P, and T_p is the number that the points around P change from 0 to 1 in clockwise.

P_7	P_8	P_1
P_6	P	P_2
P_5	P_4	P_3

Figure 5. Pixels of the center pixel P.

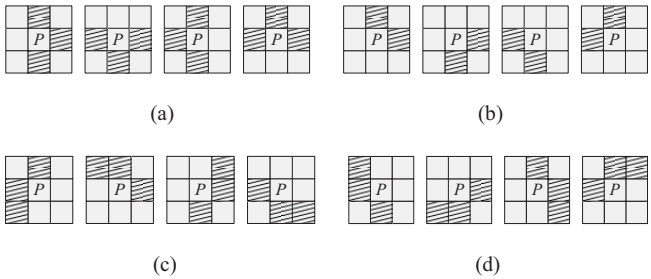


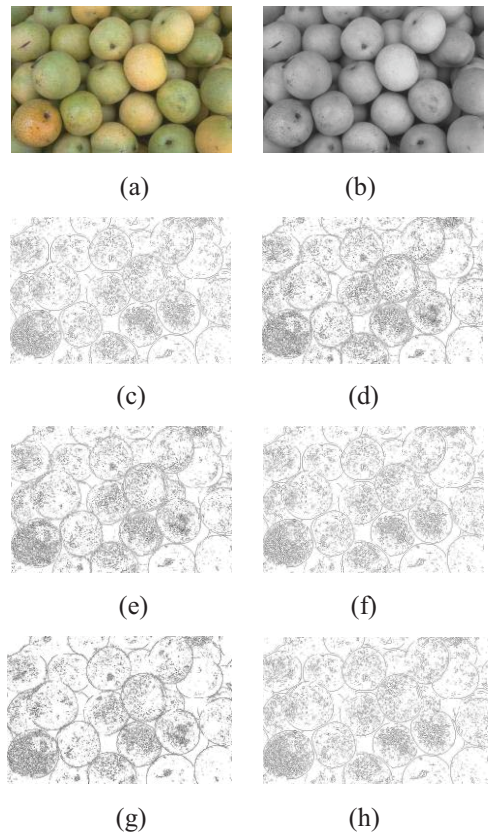
Figure 6. Four groups of thinning structure elements. Left to right: (a) A_1, A_2, A_3, A_4; (b)B_1, B_2, B_3, B_4; (c)C_1, C_2, C_3, C_4; (d)D_1, D_2, D_3, D_4.

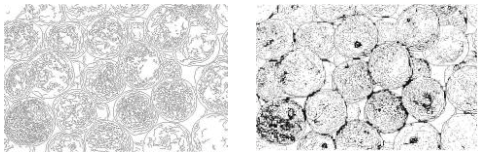
The general judging condition of the proposed edge thinning algorithm is shown in (11).

$$P = \begin{cases} 1 & else\ other \\ 0 & (\cap_{i=1}^{4} !A_i \ and \ \cap_{i=1}^{2} J_i) \ or \\ & (\cup_{i=1}^{4} B_i) \ or \ (\cup_{i=1}^{4} C_i) \ or \ (\cup_{i=1}^{4} D_i) \end{cases} \quad (11)$$

III. Experiments

For comparison, we conduct experiments on several existing edge detection methods with the pear image in Fig. 7 and the experimental result shown that the output of the proposed method is better than other methods in some cases. From the results demonstrate in Fig. 7, the advantage of the proposed method is very obvious. The edges of object in Fig.7.(J) is evidently abundant than the edge image which produced with Laplace operator, Sobel operator, Prewitt operator, Marr-hildreth operator, Roberts operator and Zero-cross operator.

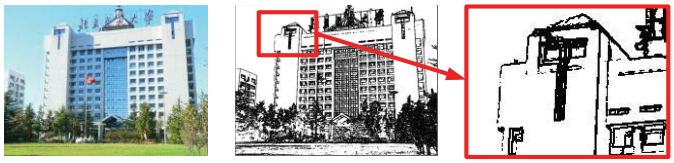




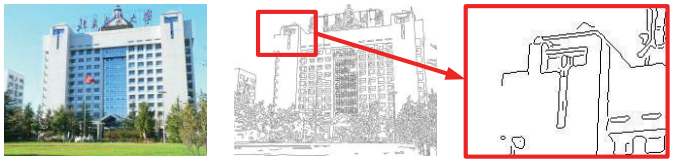
(i) (j)

Figure 7. (a)original pear image, (b)gray image, (C)result with Laplace operator with adaptive threshold, (d)result with Sobel operator with fixed threshod $T = 5$, (e)result with Prewitt operator ($T = 5$), (f)result with Marr-hildreth operator with adaptive threshold, (g)result with Roberts operator with ($T = 5$), (h)result with Zero-cross operator with adaptive threshold, (i)result with Canny operator with two adaptive threshod $T_1 = 9$ *and* $T_2 = 22$, (j)result of proposed method with adaptive threshold.

When compare to the Canny operator, it is hard to confirm which method is better in Fig.7. So, we have collected a group of image samples for outdoor scenario from Beijing Jiaotong University (BJTU). Then we conduct experiments between the proposed method and the Canny operator again. In Fig.8 and Fig.9, some edge detection results are given. The part (a) demonstrates edge detection result of the proposed method, and part (b) shows that of the Canny operator. As illustrated in the details of part (a), more edges is detected by using our method than Canny operator.



(a) Result of the proposed method



(b) Result of the Canny operator

Figure 8. The main building of BJTU.

(a) Result of the proposed method

(b) Result of the Canny operator

Figure 9. The library of BJTU.

TABLE I. DETAIL CONSUME TIME WITH ONE 320×240 RGB IMAGE

Steps	Consume Time (ms)
Smoothing	47
Color edge detection	16
Adaptive thresholding	1
Edge tinning	141

IV. CONCLUSIONS AND FUTRUE WORK

In this paper, we present a color edge detection approach and we implement the proposed method on one PC with Intel Pentium Dual CPU of 1.60GHz and 512MB memory. When deal with one 320×240 original RGB image, the consume time of our general method is about 205ms and the detail consume time as shown in Table I. Experiment results show that the proposed approach is stable and effective. However, the proposed method may not be applied for the real-time application task that the edge image is required in a very short time. So, our next goal will be improving each step of our method to subdue the complexity and the computing time.

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